FIGURE 11.1a

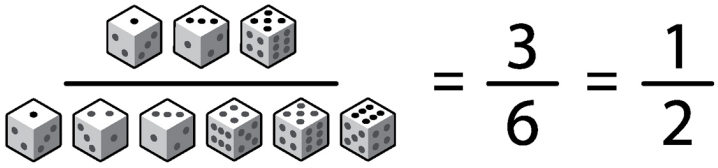


FIGURE 11.1b

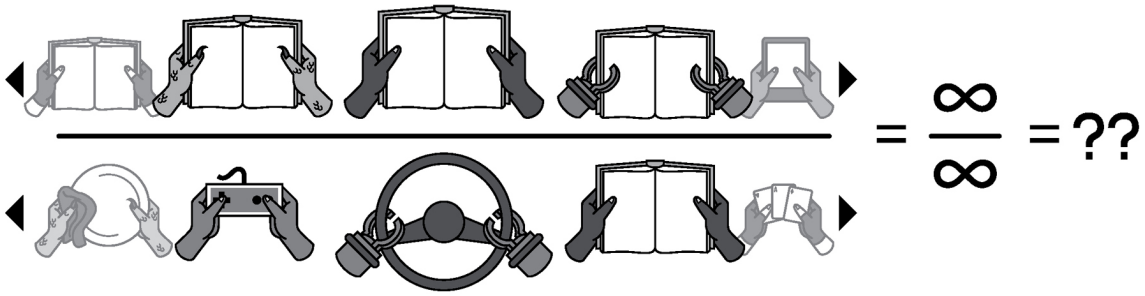


Figure 11.1. (a) Probabilities are relatively easy to calculate for dice and other situations where there are only a finite number of outcomes. The odds of rolling an odd number with an ordinary six-sided die are three out of six, or one-half. (b) Probabilities are much more difficult to calculate in an infinite multiverse. In the many-worlds interpretation, what are the odds that a randomly chosen version of you is reading this book right now?

The answer, or at least an answer, is that probability shows up in the many-worlds interpretation because we're hopelessly lost. Although the universal wave function obeys the Schrödinger equation and splits in a deterministic way, we don't know where in that huge and complicated wave function we are. We know that we're in only one branch of the universal wave function—but which one? After all, there are many copies of each of us scattered across a multitude of quantum worlds, each only slightly different, so it's not immediately obvious which world we're in. In particular, after conducting a quantum experiment, we know that we're in one and only one of the several worlds that the universe split into after the experiment concluded. But we can't know which of those worlds we're in without looking at the outcome of the experiment—we can't tell just by looking around us, because all of those universes look identical otherwise. The best we can do is to use the mathematics of quantum physics to say